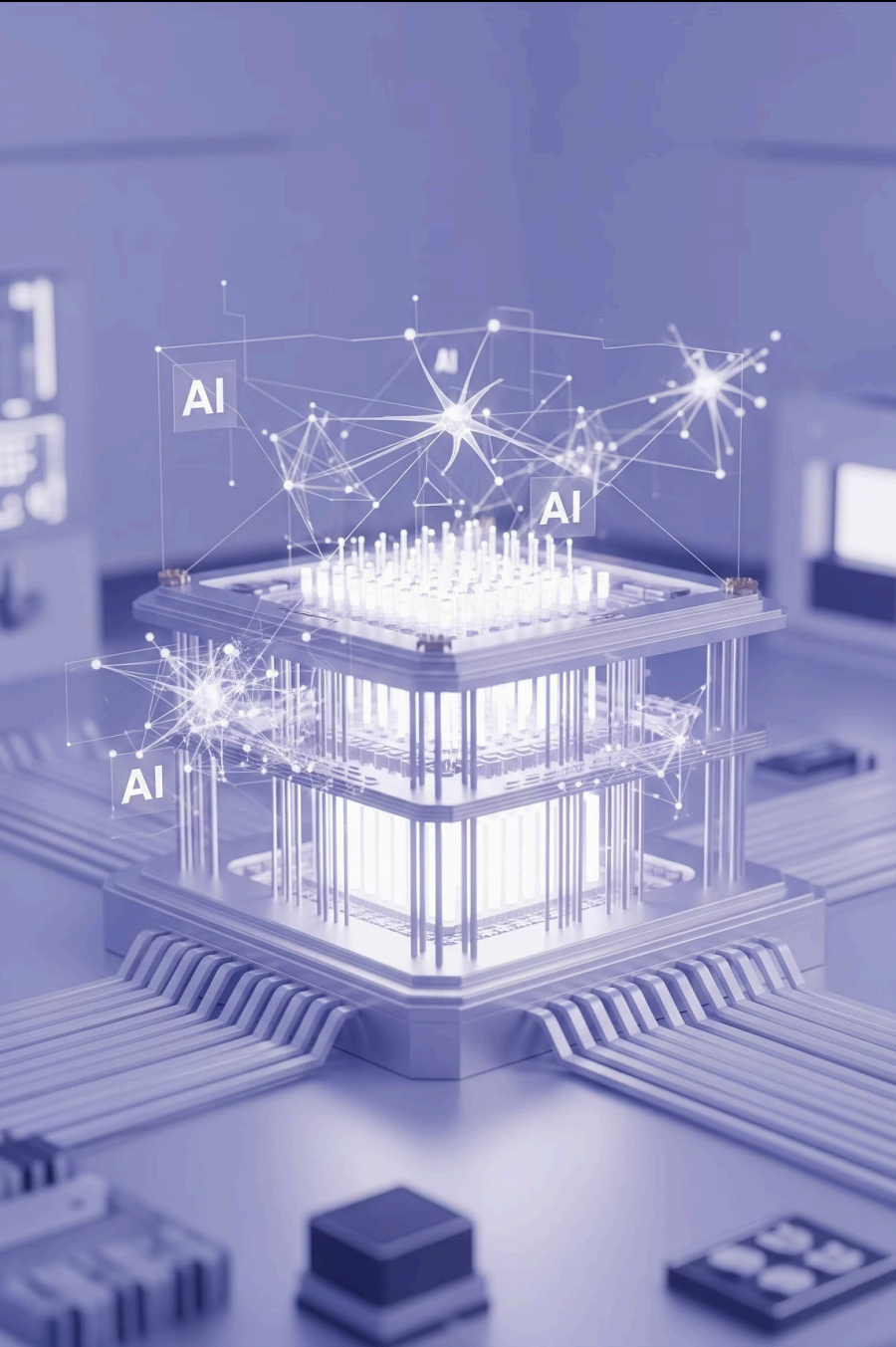




AI + Quantum Computing Infrastructure

Exploring the next frontier where quantum computation meets artificial intelligence - a convergence that promises to redefine the boundaries of computational power and problem-solving capabilities.

Why Combine AI + Quantum?

1

Exponential Compute Demands

AI's insatiable appetite for computational power finds a natural partner in quantum's exponential processing capabilities

2

Complex Problem Solving

Quantum excels at optimization and simulation problems that are computationally intensive for classical systems

3

Error Management

AI provides sophisticated tools to manage noisy quantum systems through error correction and calibration

The fusion of these technologies creates powerful hybrid systems poised to revolutionize scientific discovery and infrastructure design

Quantum Computing Fundamentals

Core Concepts

- Qubits replace classical bits, enabling **superposition** and **entanglement**
- Breakthrough algorithms: Shor's (factoring), Grover's (search), VQE, QAOA
- Current limitation: **noise + small qubit counts**

Today's quantum systems require sophisticated **classical + AI infrastructure** to reach their potential



Infrastructure for Quantum–AI Hybrid Systems

Quantum Hardware

Diverse physical implementations including superconducting circuits, trapped ions, and photonic systems

Classical HPC

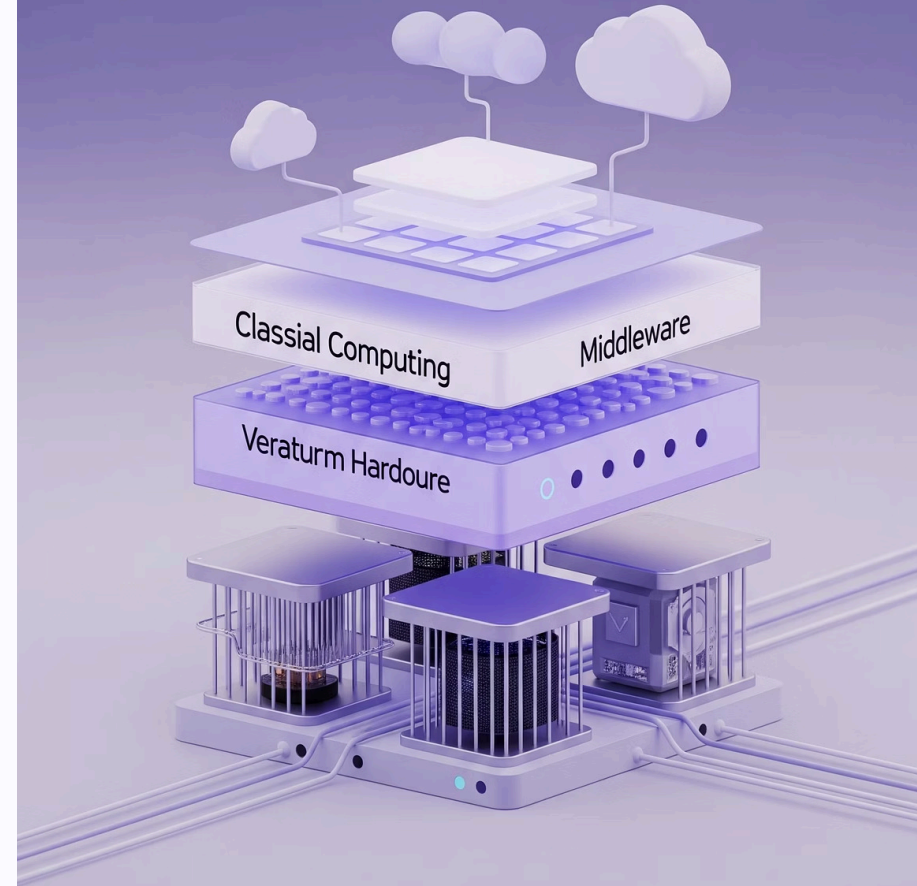
GPUs/TPUs power AI components and support hybrid algorithmic execution

Middleware

Software frameworks like Qiskit, PennyLane, and Cirq bridge quantum-classical divide

Cloud Platforms

AWS Braket, Azure Quantum, and IBM Quantum provide accessible quantum computing resources

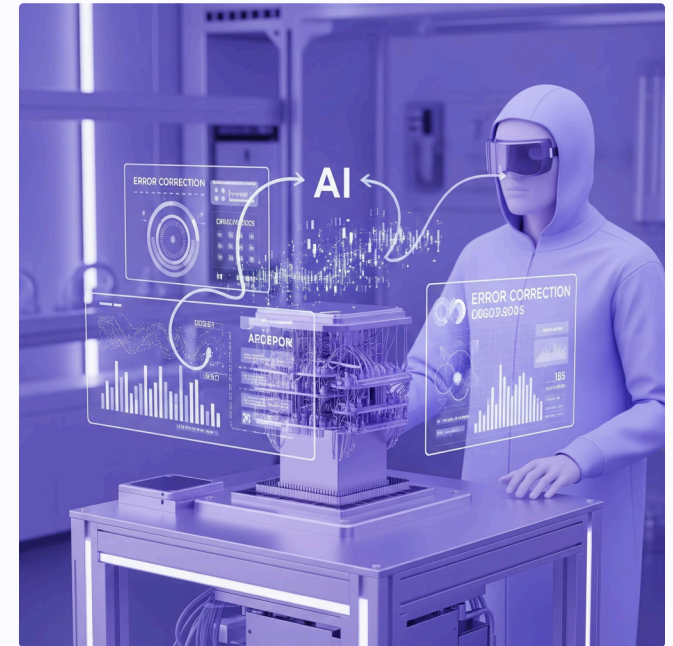


AI for Quantum Systems

Artificial intelligence is already enhancing quantum computing capabilities:

- Neural networks calibrate quantum circuits with precision
- Deep reinforcement learning optimizes error correction and qubit scheduling
- ML models predict noise patterns and optimize quantum gate operations

These advancements help extend the **NISQ (Noisy Intermediate-Scale Quantum) era** while we await fault-tolerant systems



Quantum for AI Workloads



Quantum ML (QML)

Quantum-enhanced kernel methods and Variational Quantum Classifiers promise theoretical advantages



Combinatorial Optimization

Supply chain logistics, portfolio management, and resource allocation benefit from quantum approaches

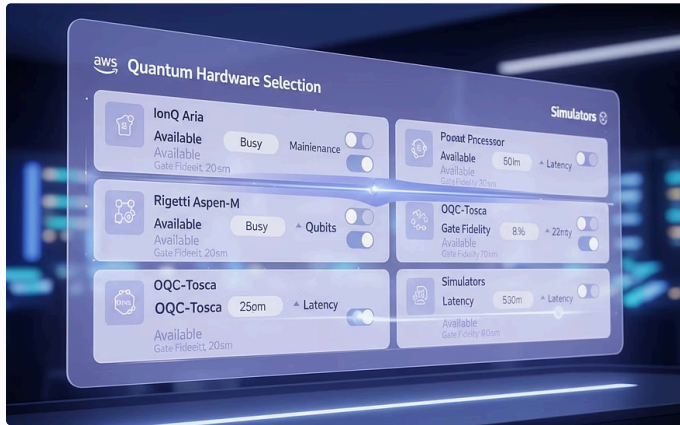


Generative Models

Quantum Boltzmann machines and quantum GANs represent emerging quantum-native architectures

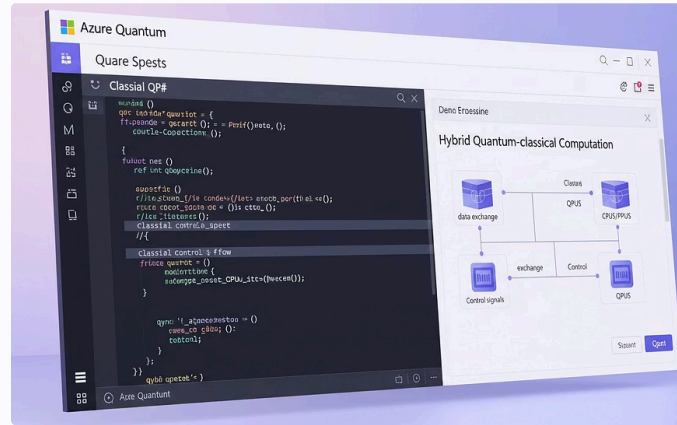
⚠ These applications remain **experimental and limited to small-scale implementations** due to current hardware constraints

Hybrid AI–Quantum Cloud Infrastructure



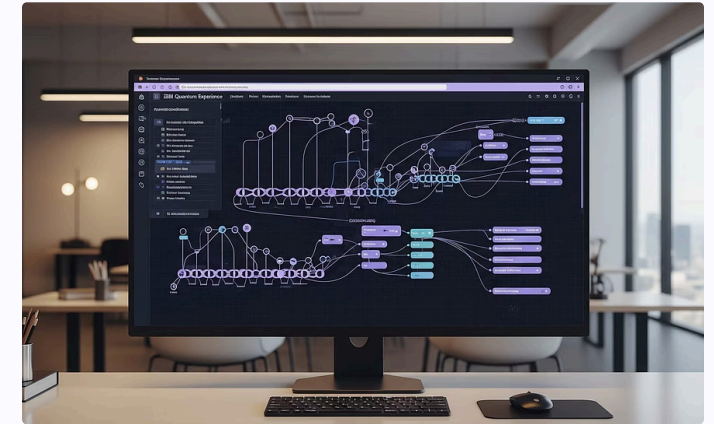
AWS Braket

Provides unified access to multiple quantum hardware providers through familiar AWS interfaces



Azure Quantum

Integrates Q# programming language with hybrid HPC capabilities



IBM Quantum

Offers cloud APIs for building quantum and classical ML pipelines

A **quantum-as-a-service model** is emerging as the predominant delivery mechanism

Data Infrastructure Challenges

- Quantum Data Encoding

Quantum algorithms require specialized data encoding schemes (amplitude encoding, angle encoding) that differ fundamentally from classical approaches

- Feature Map Bottlenecks

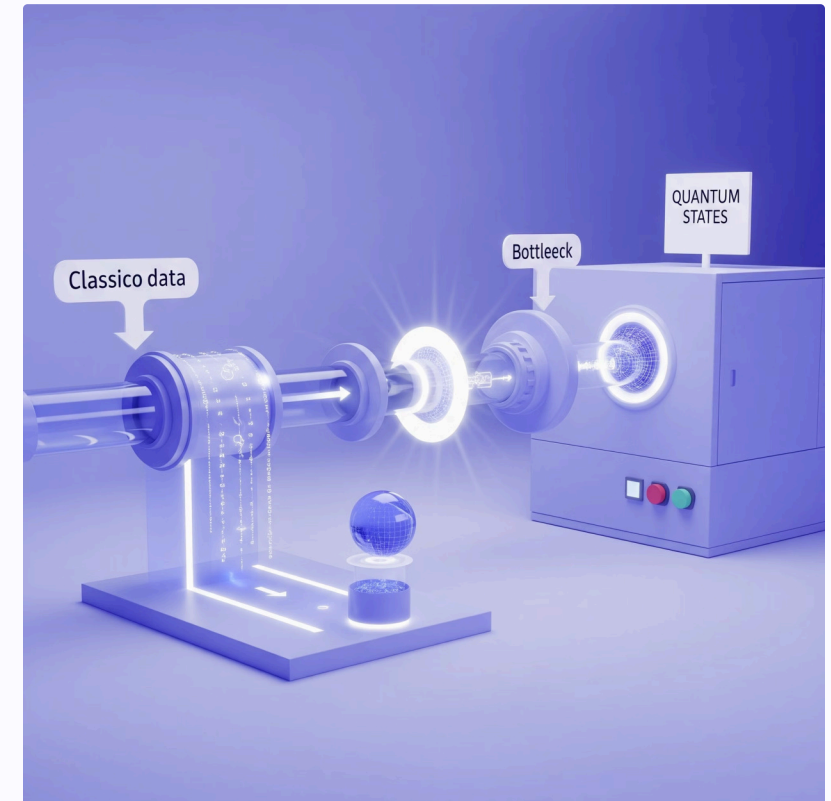
The conversion from classical data to quantum states creates performance bottlenecks in hybrid systems

- Latency Management

Hybrid systems must minimize the communication overhead between **classical** ↔ **quantum** components

- Quantum-Ready Data Lakes

Enterprise data infrastructure requires new **quantum-ready pipelines** to prepare data appropriately



Simulation Infrastructure



Bridging Today and Tomorrow

Until quantum hardware matures, sophisticated simulators serve as critical development platforms:

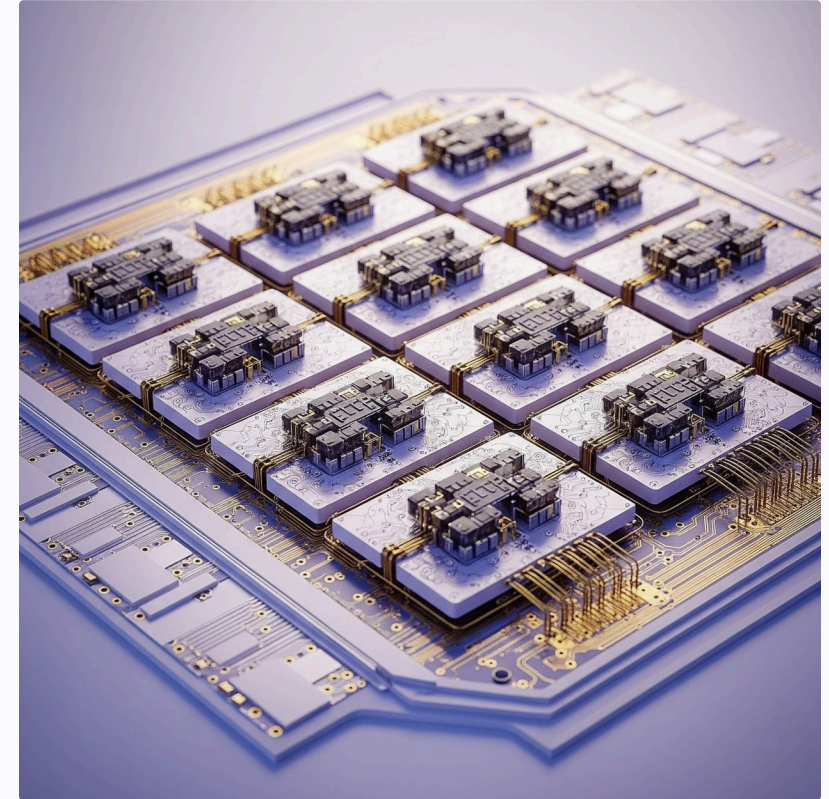
- GPU-accelerated quantum simulators support large "virtual" qubit counts
- NVIDIA cuQuantum library enables high-performance quantum circuit simulation
- Allows algorithm prototyping and testing before running on actual quantum hardware
- Essential development tool given today's limited physical qubit availability

Case Study: Google Sycamore

Quantum Supremacy Milestone (2019)

Google's landmark achievement demonstrated the potential of quantum-AI synergy:

- 53-qubit superconducting processor performed calculation that would be impractical for classical supercomputers
- Sophisticated classical control infrastructure managed qubit operations
- Machine learning algorithms stabilized quantum circuits and optimized performance
- Proved that **AI + quantum synergy** is achievable and offers computational advantages



Case Study: BMW Quantum Supply Chain



AWS Braket Implementation

- BMW utilized hybrid quantum-classical optimization for complex vehicle supply chain routing
- Combined machine learning with QAOA (Quantum Approximate Optimization Algorithm)
- Infrastructure: AWS Braket quantum backends with GPU clusters for preprocessing
- Proof-of-concept demonstrated potential cost savings in logistics operations

This industrial application highlights practical near-term value in hybrid approaches

Case Study: NVIDIA cuQuantum + PennyLane

1

GPU Acceleration

NVIDIA's cuQuantum library enables large-scale quantum circuit simulation on GPU hardware

2

ML Framework Integration

Seamless compatibility with PyTorch and TensorFlow enables quantum-classical hybrid models

3

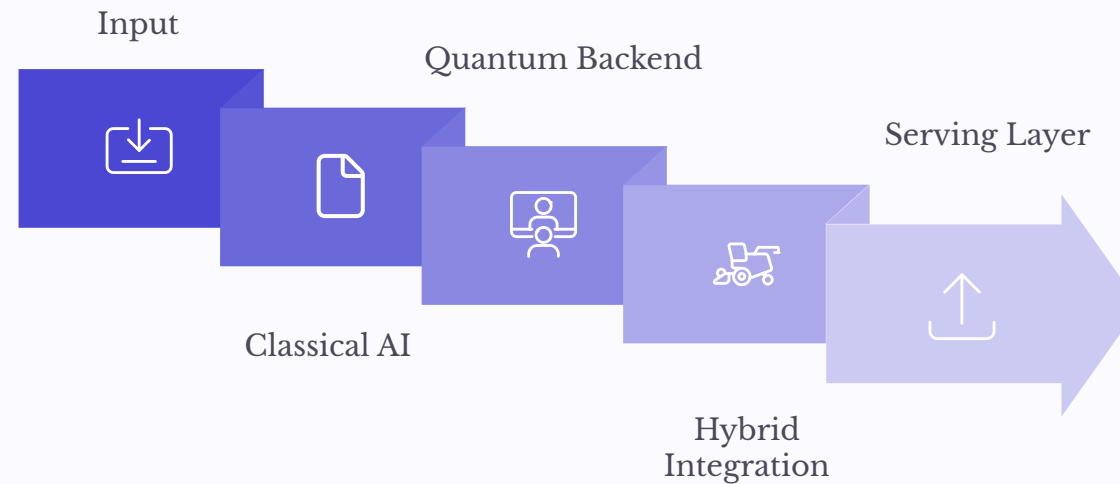
Research Enablement

Researchers can explore quantum kernels and algorithms at scales impossible on current quantum hardware

This technology stack serves as a critical bridge between today's machine learning systems and tomorrow's quantum ML capabilities



Hybrid AI–Quantum Infrastructure



Hybrid AI–Quantum infrastructure blends classical and quantum compute capabilities to deliver computational advantages beyond what either could achieve independently

Key Lessons

1 Complementary Technologies

Quantum won't replace AI infrastructure — it will augment it by addressing specific computational challenges

3 Bidirectional Benefits

AI significantly enhances quantum systems through error correction and calibration capabilities

2 Cloud-First Adoption

Hybrid cloud platforms from AWS, Azure, and IBM are leading practical adoption through accessible service models

4 New Paradigms Emerging

Quantum computing may unlock **entirely new AI approaches** in optimization and simulation domains

Strategic Investment Considerations

AI for Quantum Systems



Near-term impact: Enhancing current quantum hardware capabilities through better error correction, calibration, and control systems

Quantum for AI Workloads



Long-term breakthrough: Developing fundamentally new AI algorithms that leverage quantum advantages for previously intractable problems

Which direction would you prioritize for investment today, and why?